Architecture Checkpoint Report

Assignment 2: Architectural Evolution of nopCommerce

Group: 107
Members: Afonso Ferreira (113480), Tomás Brás (112665), Hugo Ribeiro (113402), Rodrigo Abreu (1

April 25, 2026

Contents

Chapter 1

Introduction and Scenario

1.1 Scenario Choice

This project follows **Scenario C - Omnichannel Commerce Core**. VerdeMart Retail started by using nopCommerce as its online storefront, but the business now needs more than a web shop. It wants web sales, warehouse execution, shipping, store operations, support, and customer visibility to work together as part of one commerce experience.

In this scenario, nopCommerce becomes the commerce core of that wider ecosystem. A web order should be reflected in operational systems such as warehouse and shipping, and changes that happen outside nopCommerce, such as stock or fulfillment updates, should become visible again in the customer and operator experience. This is important because those surrounding systems may be slow, stale, unavailable, or inconsistent, but the commerce experience still needs to remain useful and understandable.

1.2 Core Architectural Problem

Currently, nopCommerce mainly acts as the online shop: it owns the storefront, catalog browsing, cart, checkout, customer accounts, and order creation inside one web-centered platform. This works while the business is mostly operating through the online channel, but it does not fully address an omnichannel environment where warehouse execution, shipping, store operations, and support systems also influence the customer experience.

The core architectural problem is deciding how this online shop can evolve into a commerce core without assuming that every surrounding operational system is always fast, available, and consistent. The architecture must define what remains inside nopCommerce, what moves around it, and how the system behaves when warehouse or shipping state is delayed, stale, unavailable, or later recovered.

Chapter 2

Current-State Analysis

2.1 Overview of nopCommerce

2.2 Strengths

2.3 Limitations under Scenario

Chapter 3

Domain and Bounded Contexts

3.1 Domain Overview

3.2 Bounded Contexts

3.3 Responsibilities

3.4 Architecture Diagram

Chapter 4

Quality Attribute Scenarios

Chapter 5

Architectural Approach

5.1 Chosen Framework

5.2 Justification

Chapter 6

Target Architecture

6.1 Overview

6.2 Components

6.3 Interactions

6.4 Data Ownership

6.5 Main Architecture Diagram

Chapter 7

Architectural Decisions

7.1 ADR 1 - XXX

7.1.1 Context

7.1.2 Decision

7.1.3 Alternatives Considered

7.1.4 Consequences

7.2 ADR 2 - XXX

7.2.1 Context

7.2.2 Decision

7.2.3 Alternatives Considered

7.2.4 Consequences

7.3 ADR 3 - XXX

7.3.1 Context

7.3.2 Decision

7.3.3 Alternatives Considered

7.3.4 Consequences

Chapter 8

Risk and Validation Plan

Chapter 9

Evolution Roadmap

Chapter 10

Feasibility Spike

10.1 What Was Tested

10.2 What Worked

10.3 Evidence